PRACTICAL NO 5: Configuring a Zone-Based Policy Firewall (ZPF)

Cisco IOS® Software Release 12.4(6)T introduced Zone-Based Policy Firewall (ZFW), a new configuration model for the Cisco IOS Firewall feature set. This new configuration model offers intuitive policies for multiple-interface routers, increased granularity of firewall policy application, and a default deny-all policy that prohibits traffic between firewall security zones until an explicit policy is applied to allow desirable traffic.

Nearly all classic Cisco IOS Firewall features implemented before Cisco IOS Software Release 12.4(6)T are supported in the new zone-based policy inspection interface:

1) Stateful packet inspection
2) VRF-aware Cisco IOS Firewall
3) URL filtering
4) Denial-of-Service (DoS) mitigation

Cisco IOS Software Release 12.4(9)T added ZFW support for per-class session/connection and throughput limits, as well as application inspection and control:

1) HTTP
2) Post Office Protocol (POP3),
3) Internet Mail Access Protocol (IMAP),
4) Simple Mail Transfer Protocol/Enhanced Simple Mail Transfer Protocol (SMTP/ESMTP)
5) Sun Remote Procedure Call (RPC)
6) Instant Messaging (IM) applications:
   i) Microsoft Messenger
   ii) Yahoo! Messenger
   iii) AOL Instant Messenger
7) Peer-to-Peer (P2P) File Sharing:
   i) Bittorrent
   ii) KaZaA
   iii) Gnutella
   iv) eDonkey

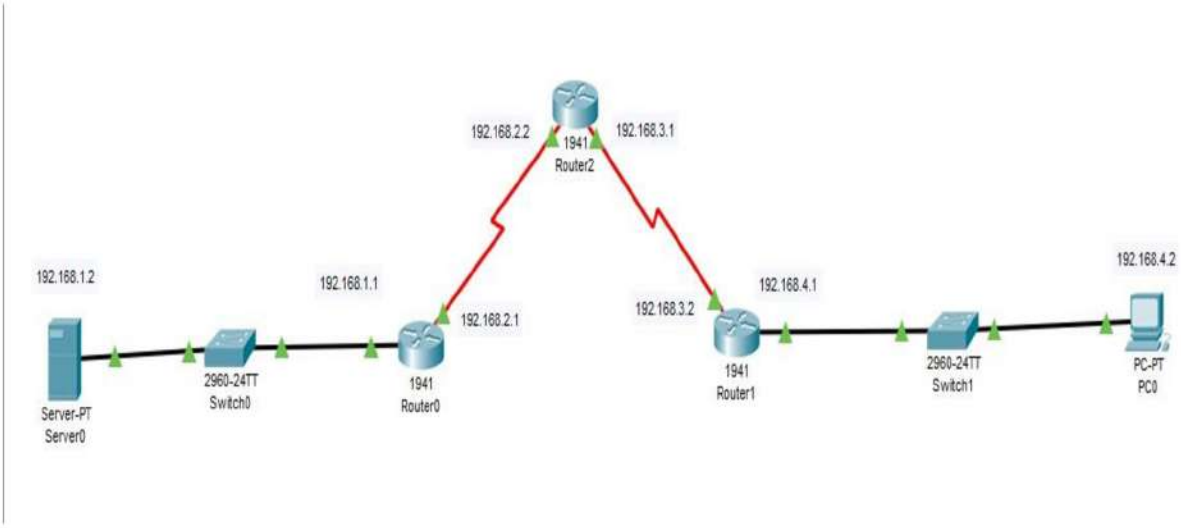
Cisco IOS Software Release 12.4(11)T added statistics for easier DoS protection tuning.

Some Cisco IOS Classic Firewall features and capabilities are not yet supported in a ZFW in Cisco IOS Software Release 12.4(15)T:

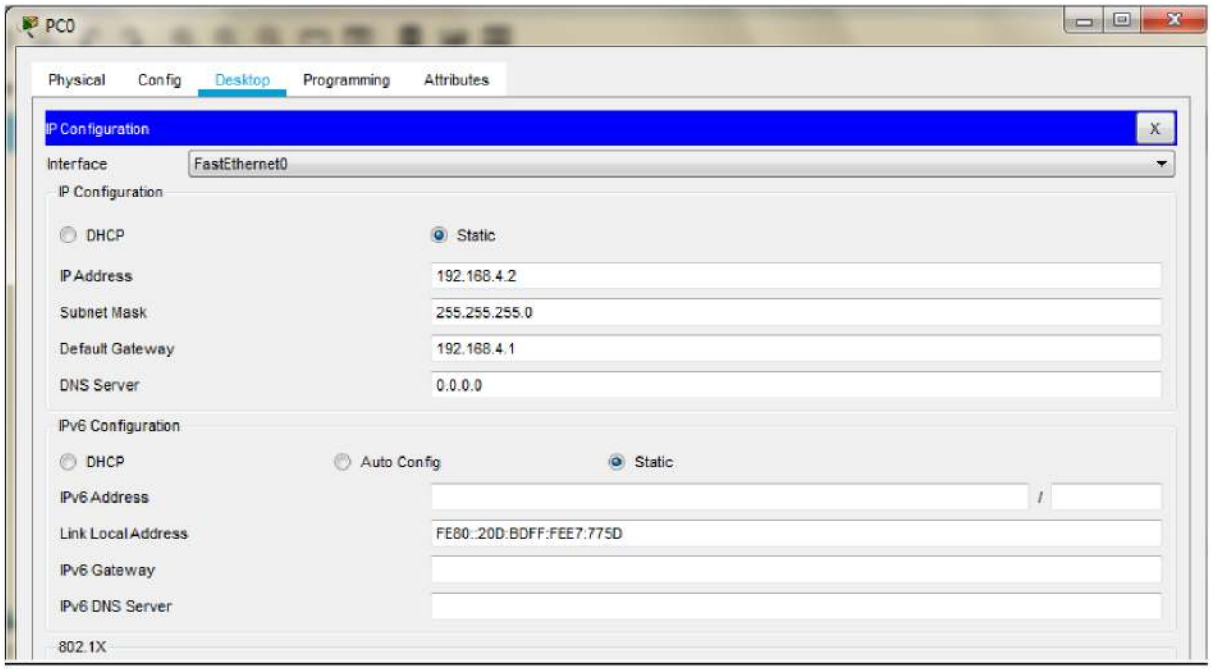
i) Authentication proxy
ii) Stateful firewall failover
iii) Unified firewall MIB
iv) IPv6 stateful inspection
v) TCP out-of-order support

ZFW generally improves Cisco IOS performance for most firewall inspection activities. Neither Cisco IOS ZFW or Classic Firewall include stateful inspection support for multicast traffic.

We use the following Topology for the current case

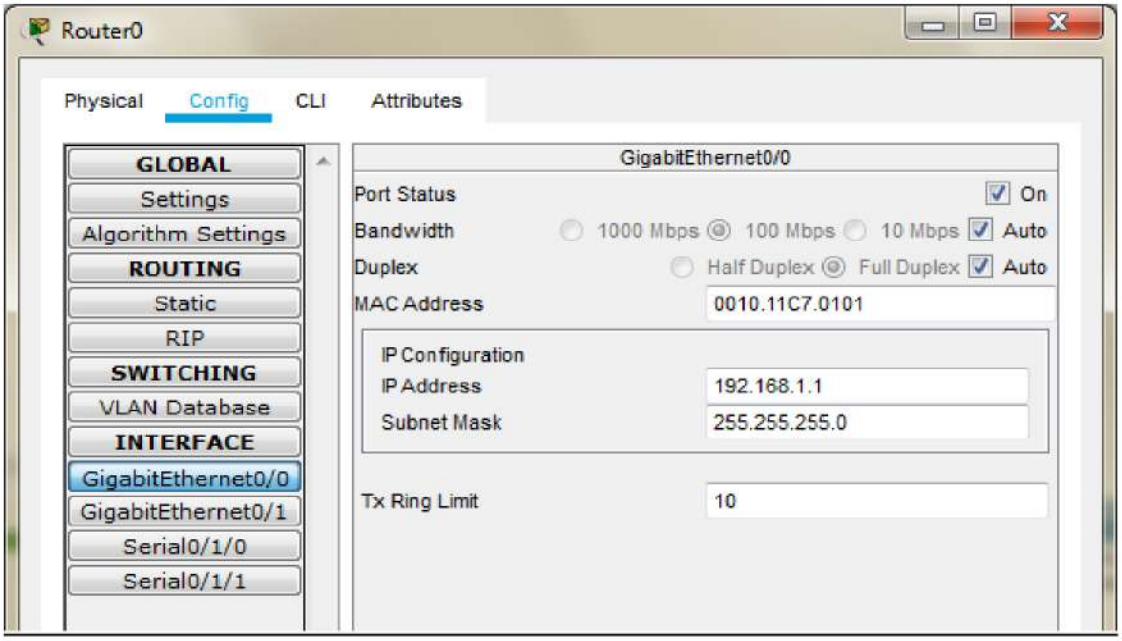


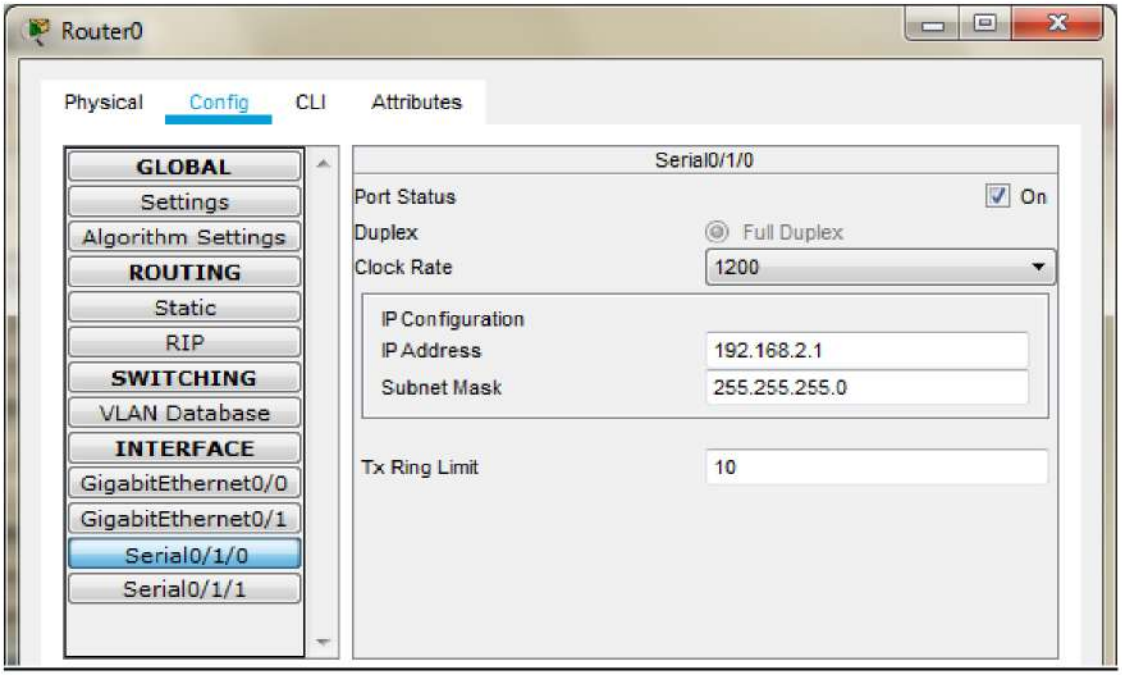
Configuring PC0



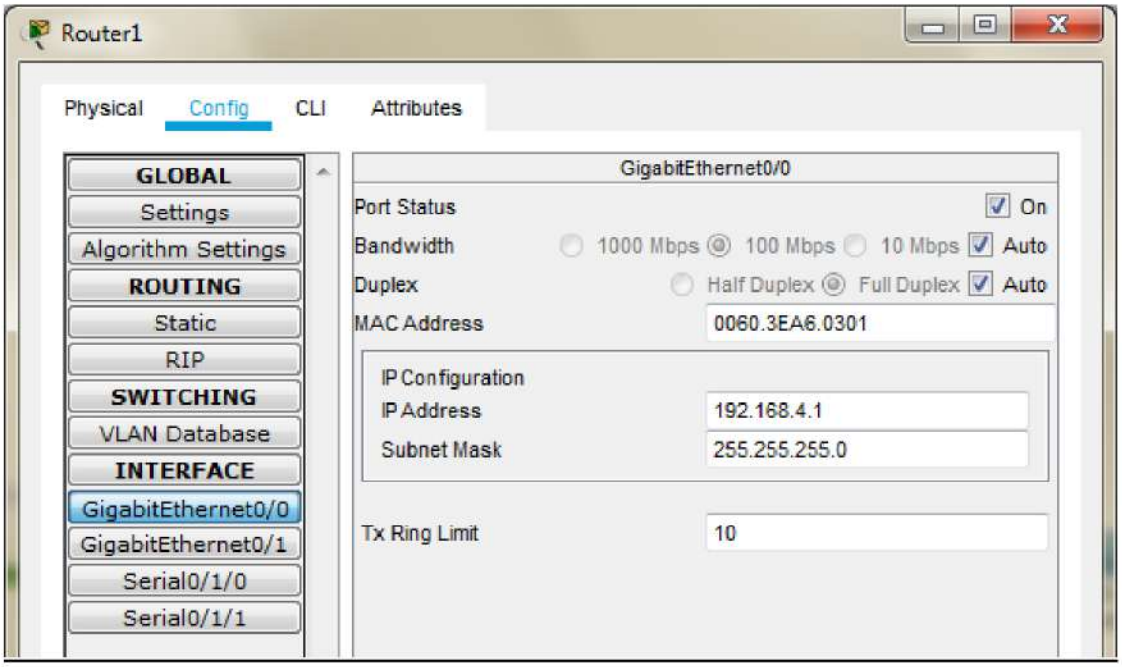
Serial Interface must be added in each Router before configuring it

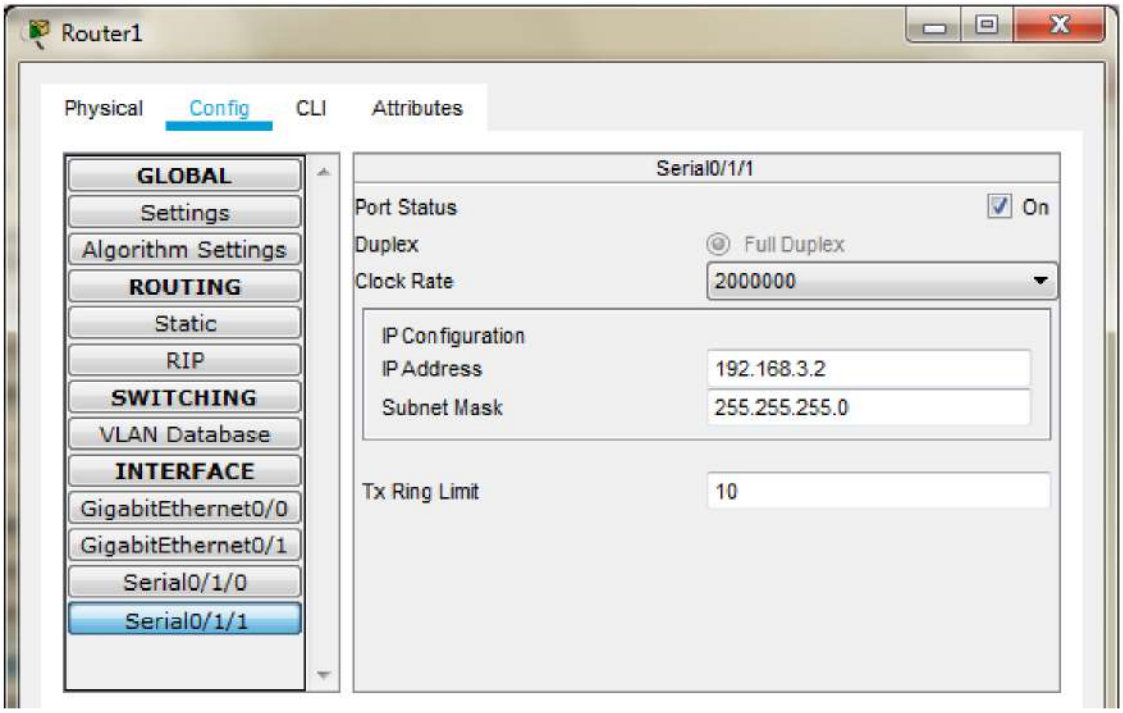
Configuring Router0



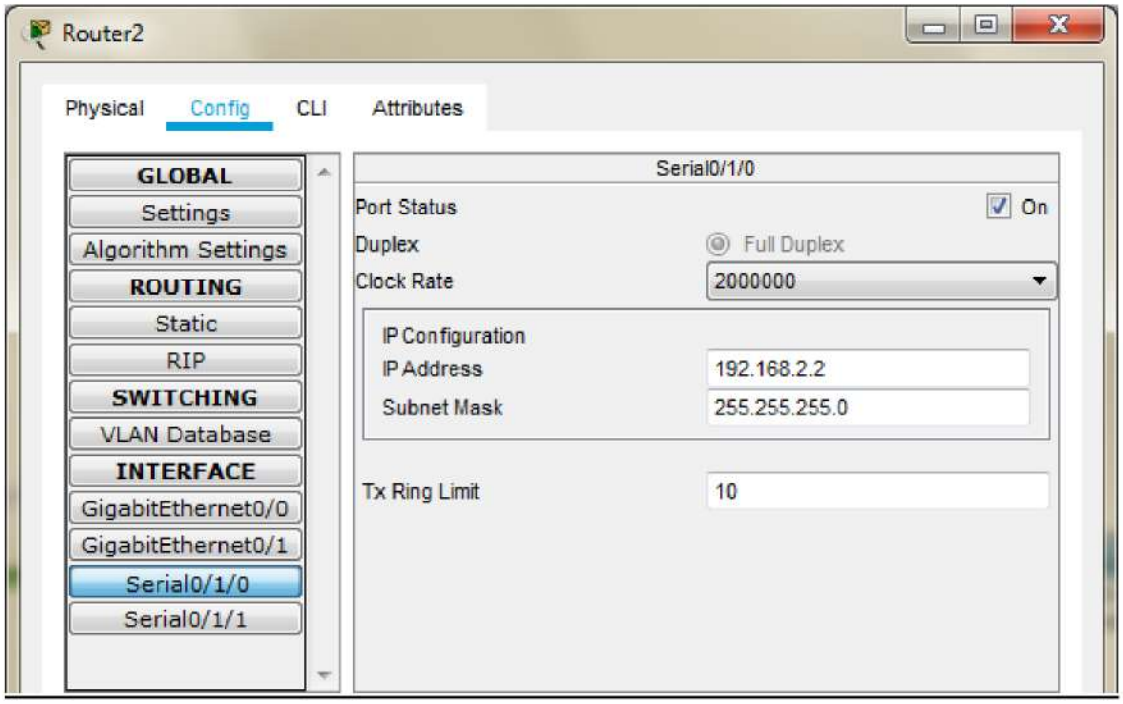


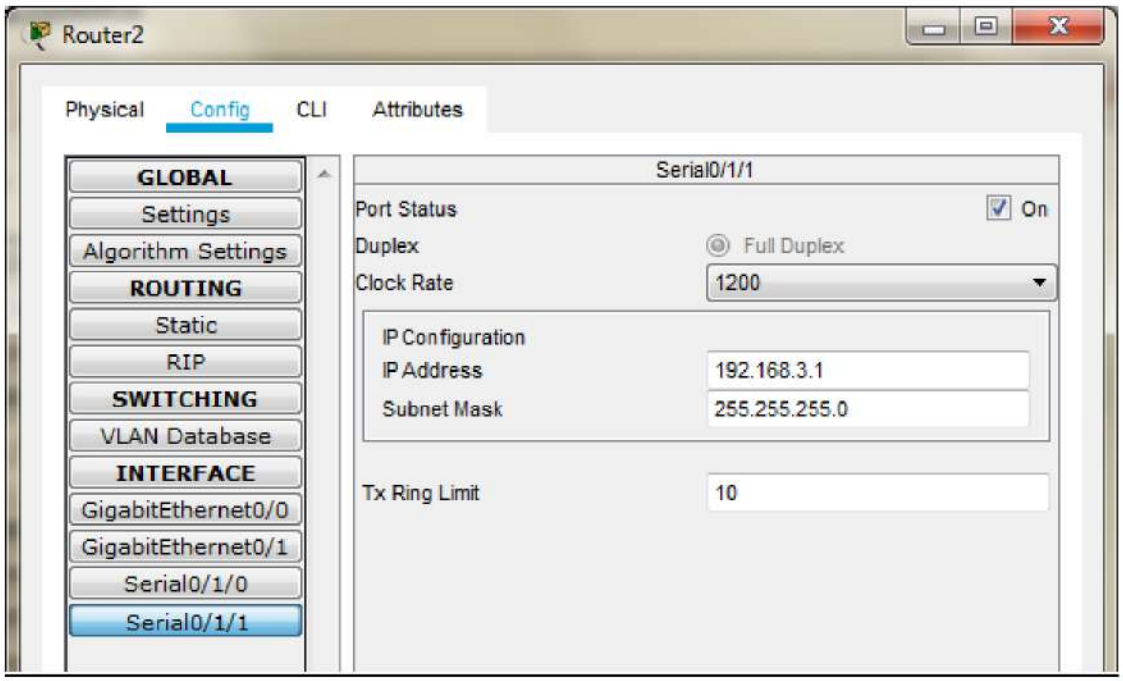
Configuring Router1



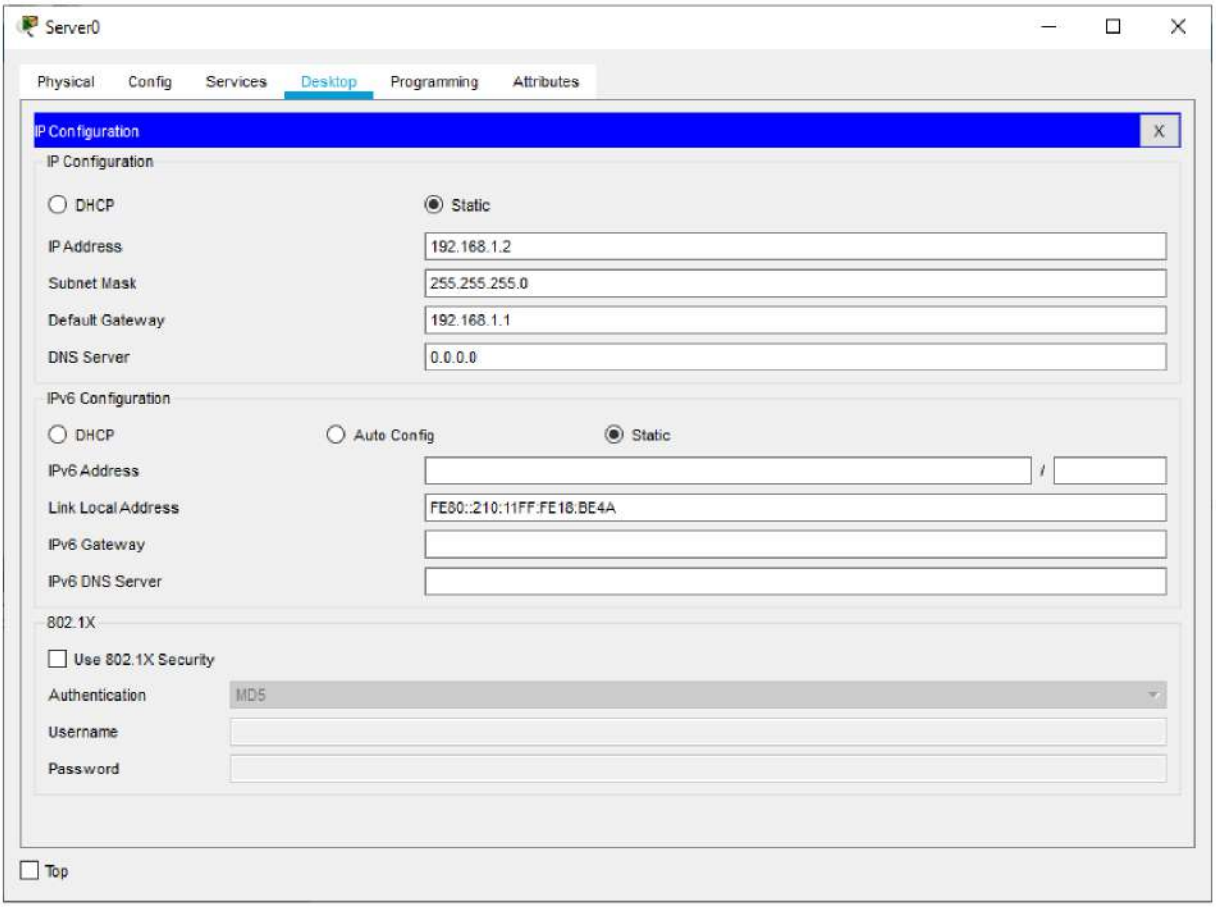


Configuring Router2





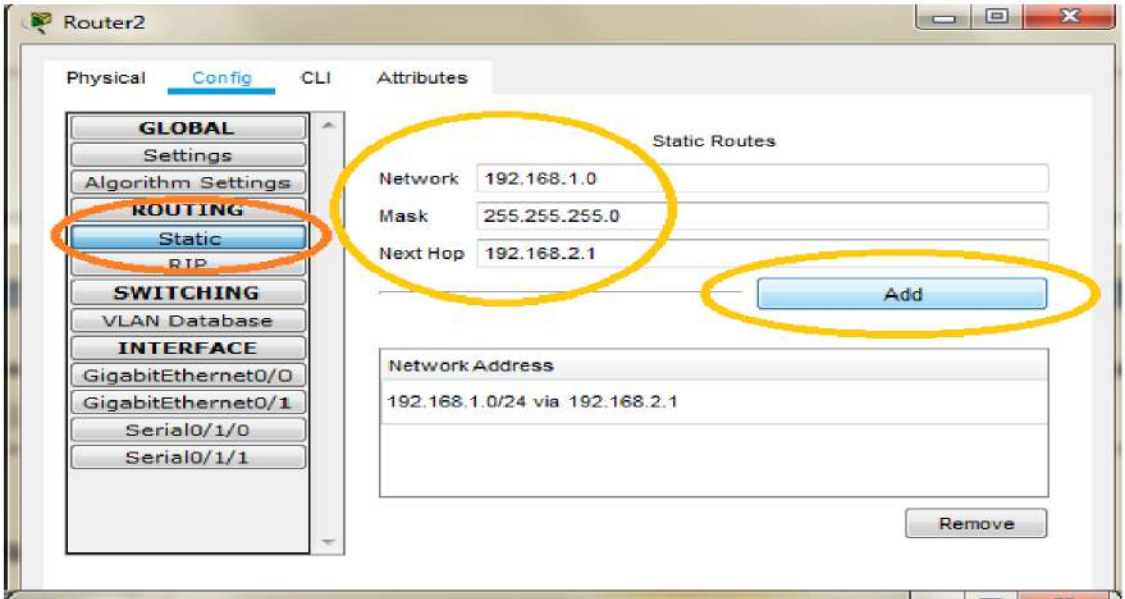
Configuring Server0

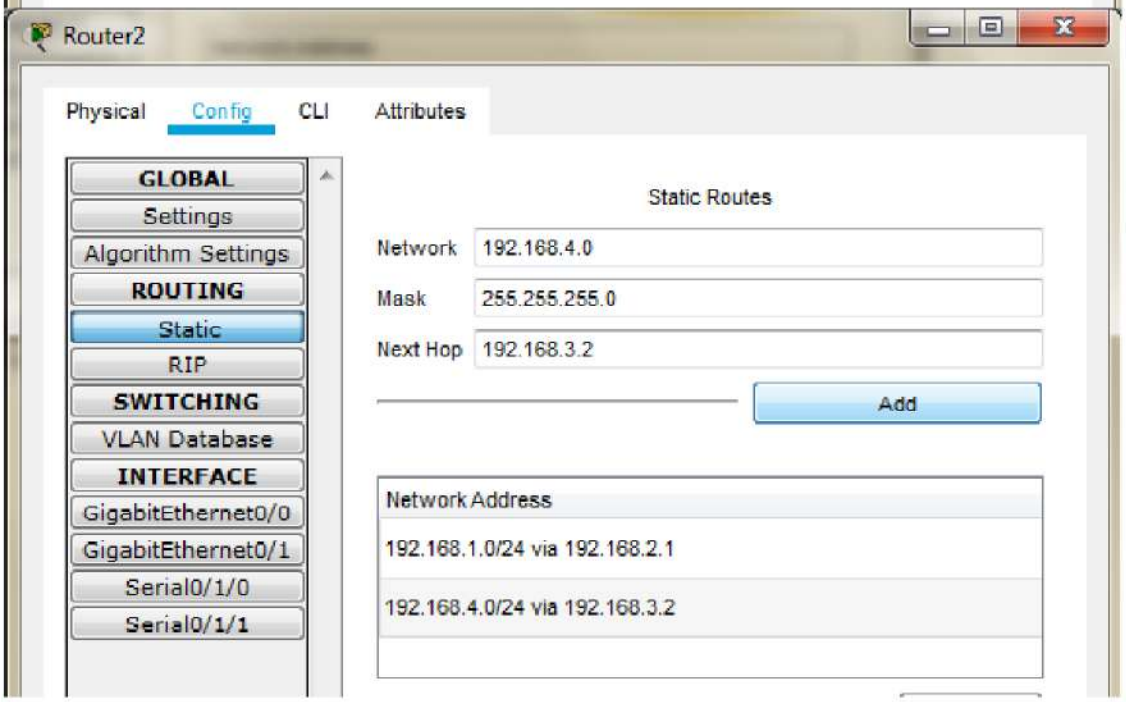


Part 1: Static Routing

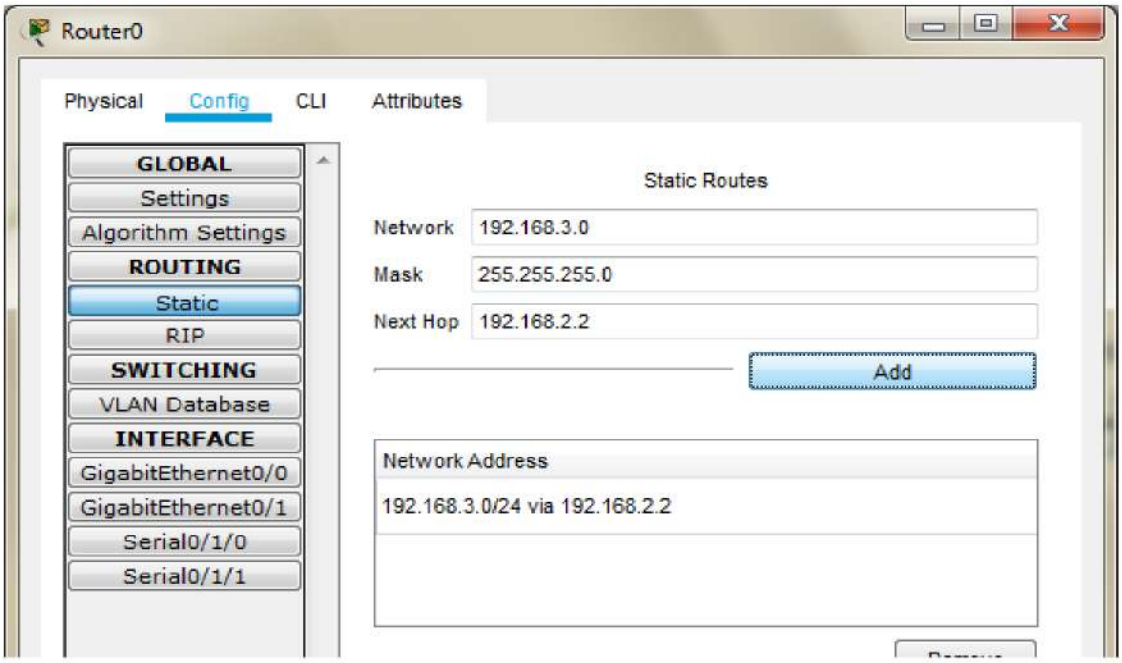
Static Routing is done using the following procedure for each Router

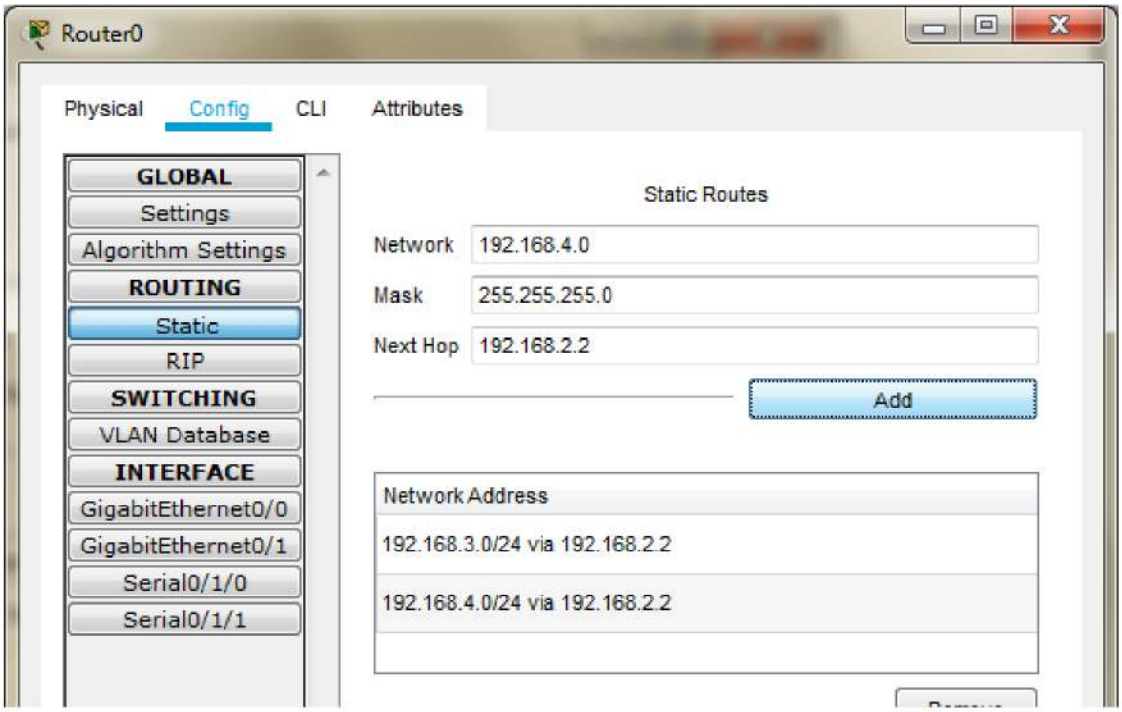
Router 2: Add the following Routes in the Static mode



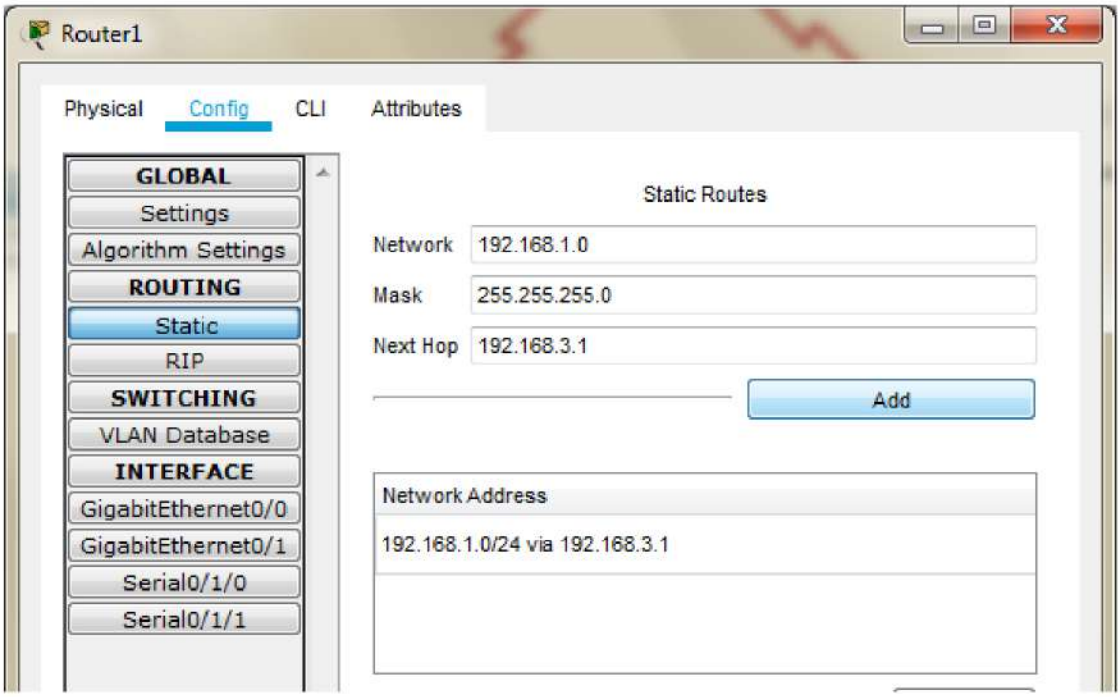


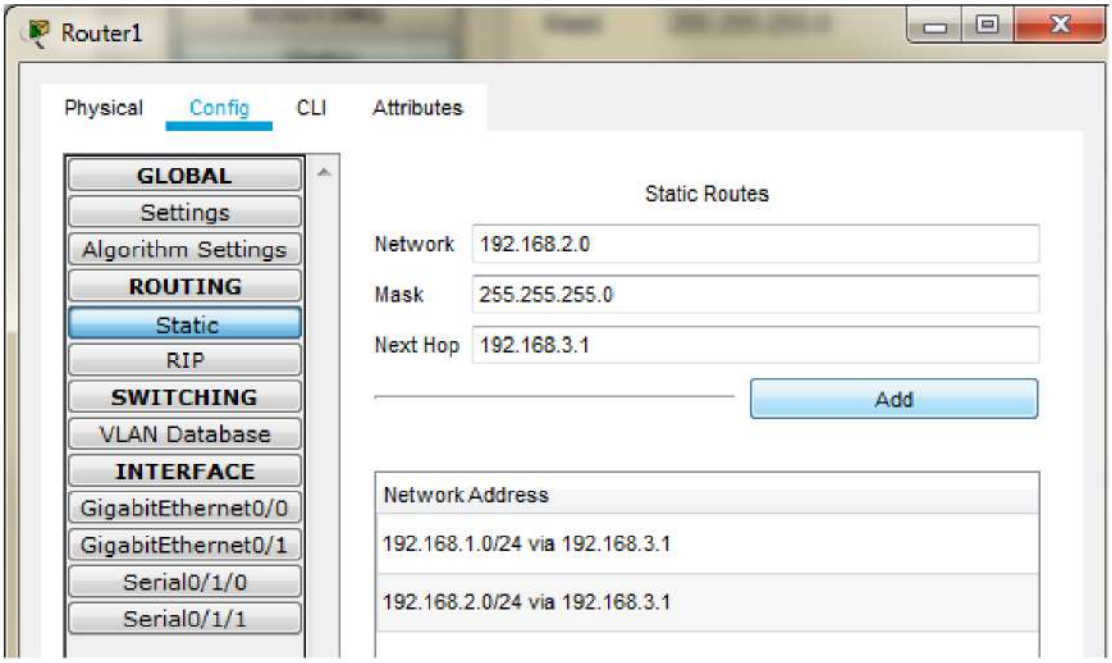
Router 0: Add the following Routes in the Static mode



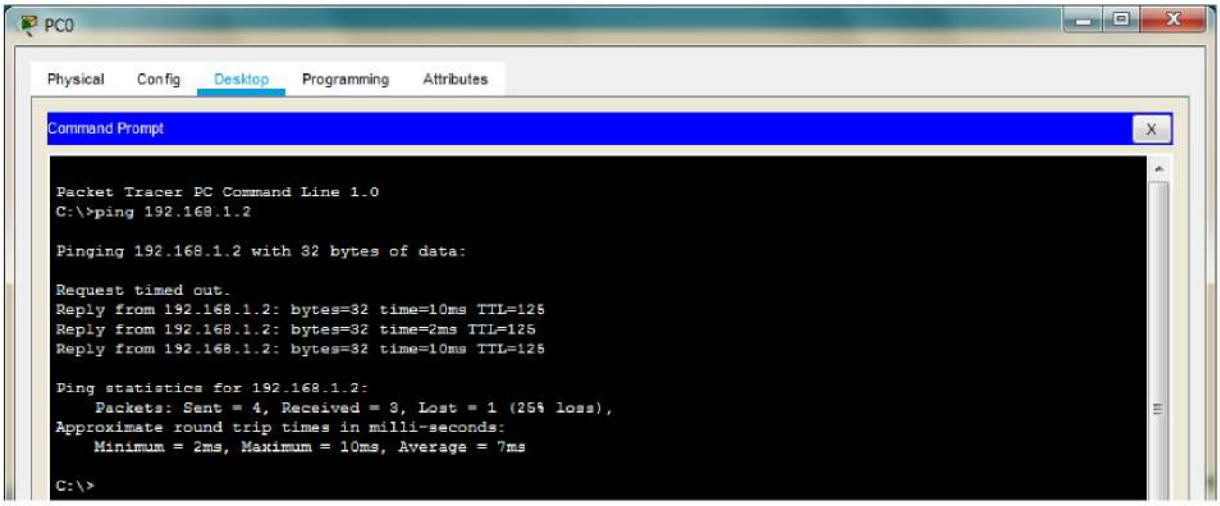


Router 1: Add the following Routes in the Static mode





Now we check the connectivity by pinging the Server from the PC

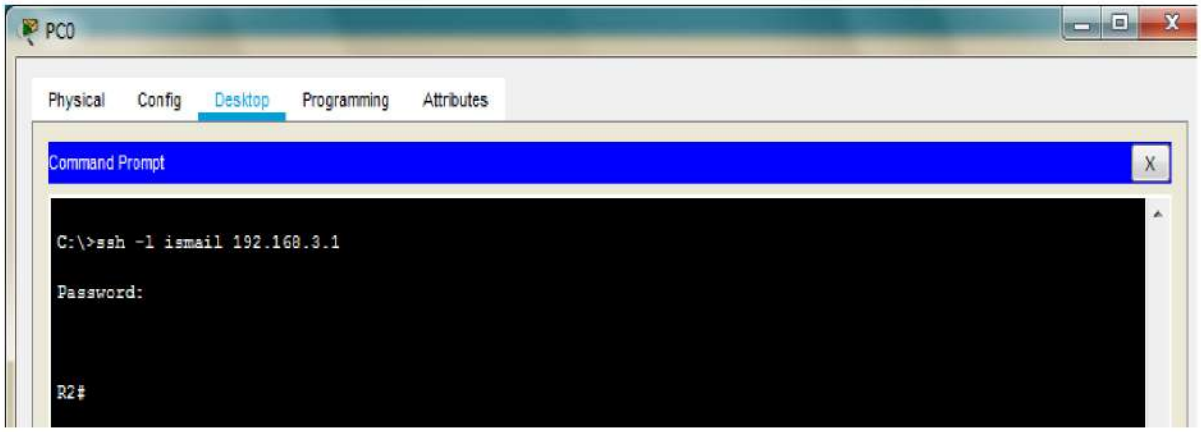


Part 2: Configuring SSH on Router 2

Type the following commands in the CLI mode of Router2

```
Router>en
Router>enable
Router#configure terminal
Router(config)#ip domain-name ismail.com
Router(config)#hostname R2
R2(config)#cryp
R2(config)#crypto k
R2(config)#crypto key g
R2(config)#crypto key generate r
R2(config)#crypto key generate rsa

R2(config)#line vty 0 4
R2(config-line)#transport input ssh
R2(config-line)#login local
R2(config-line)#exit
R2(config)#username ismail privilege 15 password cisco
```

Now we verify the SSH using PC as follows



Next we access the web services of the Server using the web browser of PC using the following

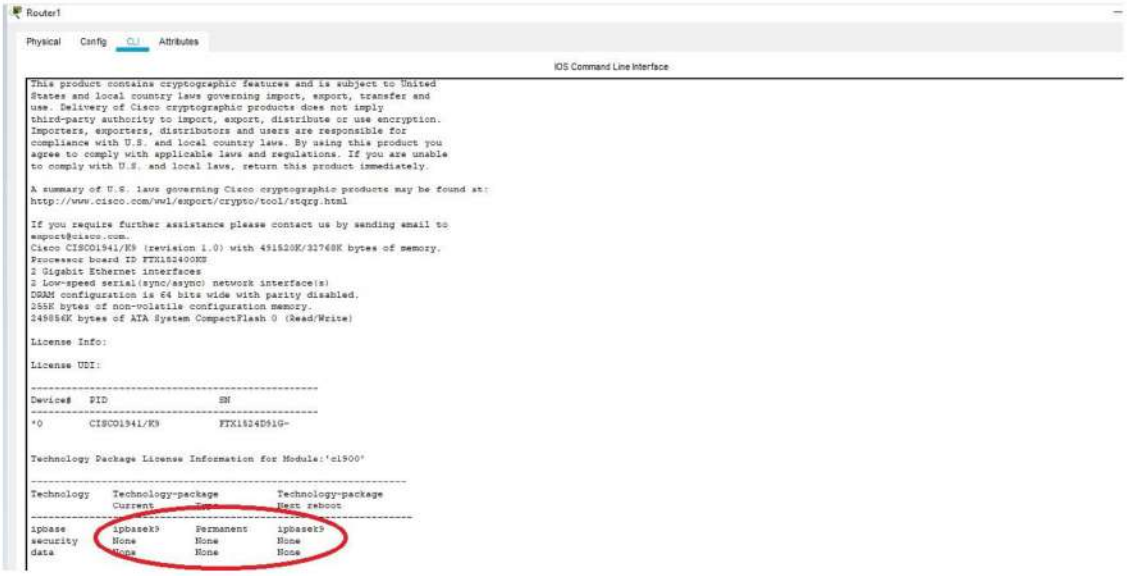
Part 3: Create the Firewall Zones on Router1

We first check the weather security package enabled or not on Router1.

Type the following command in CLI mode of Router 1

Router#show version

We will get a message informing whether the security Package is enabled or not



As seen above the security package is not enabled, to enable the security feature, type the following command in Router1

Router(config)#license boot module c1900 technology-package securityk9
Router(config)#exit
Router#
Router#reload

Router>enable
Router#
Router#show version

We will get a message informing whether the security package is enabled or not

```
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX1524D91G
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

-------------------------------------------------
Device#   PID                    SN
-------------------------------------------------
*0        CISCO1941/K9           FTX1524D91G-


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102
```

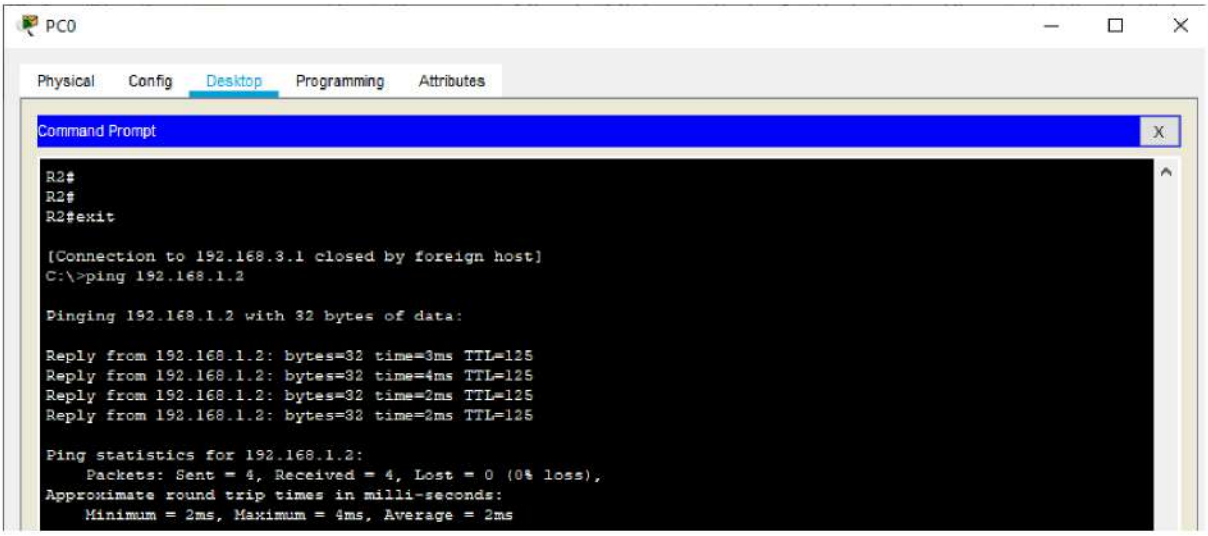
As seen above now the security package has been enabled Now type the following commands in the CLI mode of Router1

```
Router#
Router#configure terminal

Router(config)#zone security in-zone
Router(config-sec-zone)#exit

Router(config)#zone security out-zone
Router(config-sec-zone)#exit

Router(config)#access-list 101 permit ip 192.168.4.0 0.0.0.255 any
Router(config)#class-map type inspect match-all in-map
Router(config-cmap)#match access-group 101
Router(config-cmap)#exit

Router(config)#policy-map type inspect in-out
Router(config-pmap)#class type inspect in-map
Router(config-pmap-c)#inspect
Router(config-pmap-c)#exit
Router(config-pmap)#exit
Router(config)#

Router(config)#zone-pair security in-out-zone source in-zone destination out-zone
Router(config-sec-zone-pair)#service-policy type inspect in-out
Router(config-sec-zone-pair)#exit
Router(config)#

Router(config)#interface GigabitEthernet0/0
```

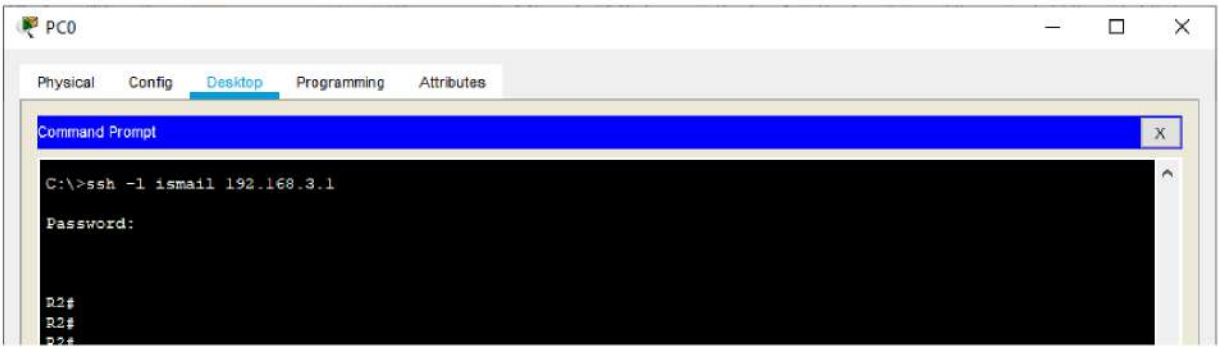
```
Router(config-if)#zone-member security in-zone
Router(config-if)#exit
Router(config)#

Router(config)#interface Serial0/1/1
Router(config-if)#zone-member security out-zone
Router(config-if)#exit
Router(config)#exit

Router#copy running-config startup-config
```

Part 4: Testing the Firewall Functionality (from in-zone to out-zone)by the following steps

Step 1: Pinging SERVER from the PC (it will succeed)



Step 2: Start an SSH session from PC to Router 2 (ip 192.168.1.2)

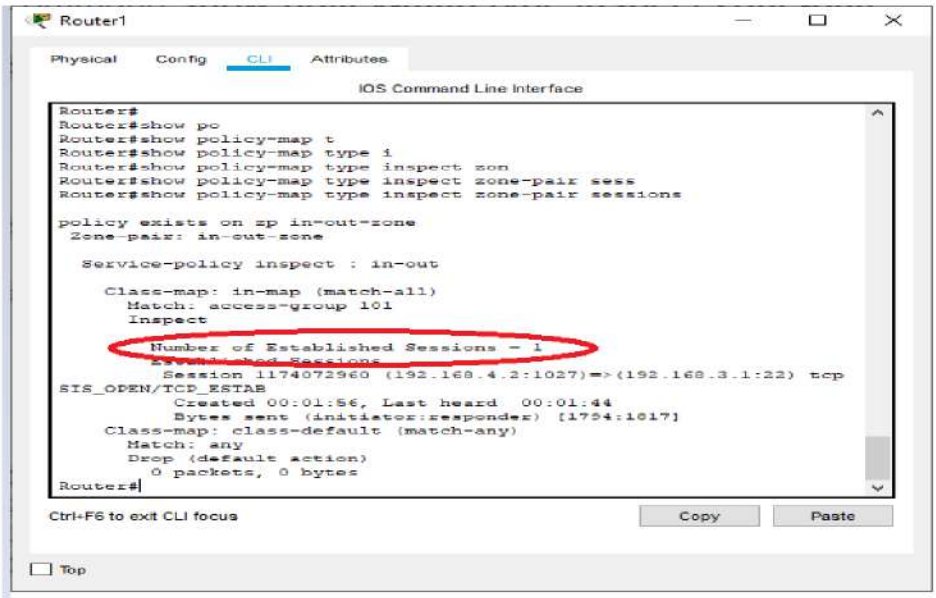


As seen above the session becomes active and we get access to Router2 (Do not exit the session and continue to Step 3)

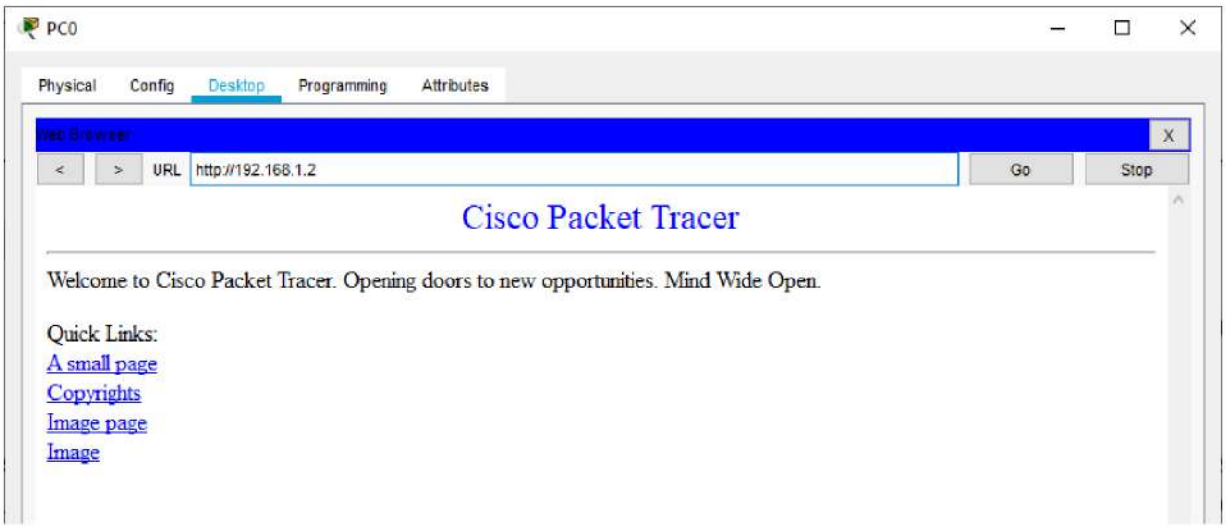
Step 3: Type the following command in the CLI mode of Router1

Router#show policy-map type inspect zone-pair sessions

We will get the following output

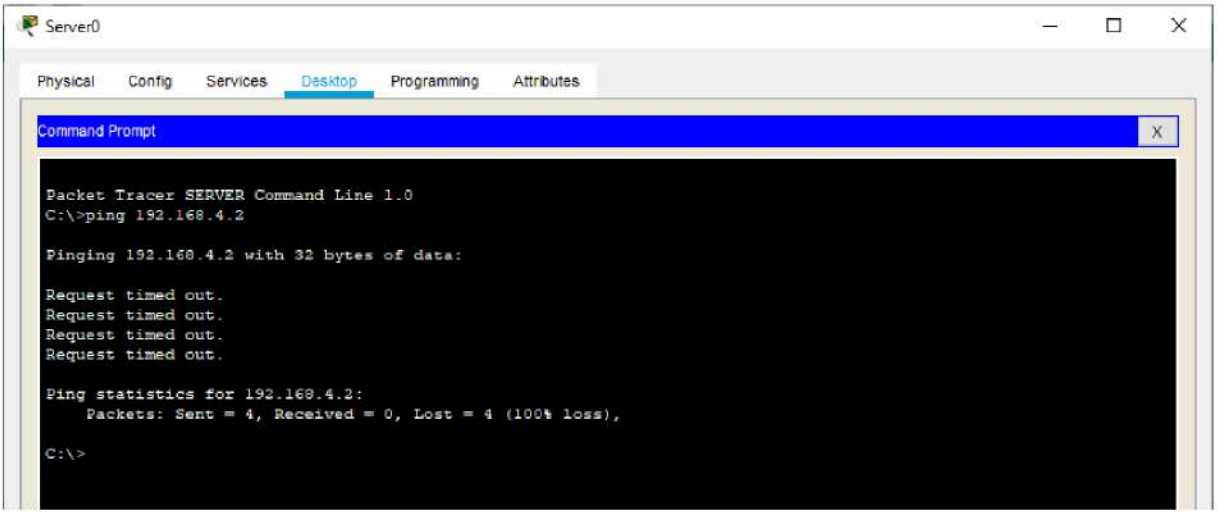


Step 4: We close the SSH connection and open the web browser and access the server address (192.168.1.2) and get the following



Part 5: Testing the Firewall Functionality (from out-zone to in-zone)by the following steps

Step 1: Ping PC0 from the SERVER (it will result in Failure)



Hence the Firewall functionality has been verified

For the video demonstration of the above practical scan the QR-code